Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 1 of 17 · Question 6 part a.iv · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

6 (a)(i)	A aorta;		1
(ii)	D vena cava;		1
(iii)	• wider lumen; • thinner walls; • valves; 	Ignore arrows Accept valves drawn either way round	1 1 1
(iv)	Any three from: (Blood vessel X) • thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure; • thinner lumen to maintain high blood pressure; OR (Blood vessel Y) - • less muscle/elastic tissue in walls so slower blood flow; • wider lumen so carries blood at lower pressure; • contains valves to prevent backflow;		Max 3

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 1 of 17 · Question 6 part a.iv · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

(b)	Any four from: • thin wall/wall one cell thick; • contains pores; • shorter diffusion pathway; • narrow lumen; • only one (blood) cell/few (blood) cells at a time can pass through;		Max 4

Total question 6 = 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 2 of 17 · Question 6 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

6 (a)(i)	A aorta;		1
(ii)	D vena cava;		1
(iii)	• wider lumen; • thinner walls; • valves; 	Ignore arrows Accept valves drawn either way round	1 1 1
(iv)	Any three from: (Blood vessel X) • thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure; • thinner lumen to maintain high blood pressure; OR (Blood vessel Y) - • less muscle/elastic tissue in walls so slower blood flow; • wider lumen so carries blood at lower pressure; • contains valves to prevent backflow;		Max 3

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 2 of 17 · Question 6 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

(b)	Any four from: • thin wall/wall one cell thick; • contains pores; • shorter diffusion pathway; • narrow lumen; • only one (blood) cell/few (blood) cells at a time can pass through;		Max 4

Total question 6 = 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 3 of 17 · Question 1 part c · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

1 (a)	pulmonary artery;		1
(b)	In the following order: • oxygenated; • lungs; • atrium; • ventricle; • body;		5
(c)	• blood vessel X has a thicker wall; • blood vessel X has a smaller lumen; • blood vessel X does not contain valves	Allow the reverse argument for blood vessel Y	3
			Total 9

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 3 of 17 · Question 1 part c · 3 marks

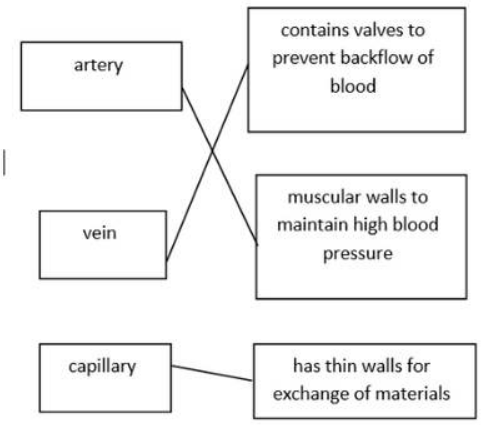
2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 4 of 17 · Question 2 part b · 2 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

2 (a) (i)	X = plasma; Y = red blood cell; Z = white blood cell/phagocyte;	Accept named phagocyte	(3)
(a) (ii)	• transports oxygen; • to body/ cells/tissues/organs;		(2)
(a) (iii)	the patient has an infection/leukaemia/allergy/ blood cancer, inflammation;	Accept pathogens in body	(1)
(b)		Reject more than one line from each blood vessel 3 correct=2marks 1 or 2 correct= 1mark	(2)
(c) (i)	left ventricle;		(1)
(c) (ii)	• into right ventricle; • through pulmonary artery; • to the lungs; • (from the lungs) into the pulmonary vein;		(4)

Total for Question 2 = 13 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 5 of 17 · Question 6 part b · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6 (a)(i)	A aorta;		1
(ii)	D vena cava;		1
(iii)	• wider lumen; • thinner walls; • valves; 	Ignore arrows Accept valves drawn either way round	1 1 1
(iv)	Any three from: (Blood vessel X) • thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure; • thinner lumen to maintain high blood pressure; OR (Blood vessel Y) - • less muscle/elastic tissue in walls so slower blood flow; • wider lumen so carries blood at lower pressure; • contains valves to prevent backflow;		Max 3

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 5 of 17 · Question 6 part b · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

(b)	Any four from: • thin wall/wall one cell thick; • contains pores; • shorter diffusion pathway; • narrow lumen; • only one (blood) cell/few (blood) cells at a time can pass through;		Max 4

Total question 6 = 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 6 of 17 · Question 4 part d · 2 marks

2020 JAN · Paper 01 · find part d in this mark scheme extract

Question number				
4	(a)	• C (left) atrium(1) • D (left) ventricle (1)	Maximum of one mark if right atrium/ventricle given	2
	(b)	Any four from the following: • (blood vessels) A and F are arteries (1) • A is the aorta/F is the pulmonary artery (1) • arteries carry blood away from the heart (1) • blood vessels B and E are veins (1) • B is the vena cava/E is the pulmonary vein (1) • veins carry blood towards the heart (1)		4
	(c)	the heart pumps blood around the body/liver detoxifies the blood/involved in digestion / homeostasis/liver not muscular/liver has no chambers (1)		1
	(d)	(capillaries) exchange of materials(between the blood and body tissues/cells)/allows diffusion (1) named substance (1)		2

Total for Question 4 = 9 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 7 of 17 · Question 8 part a.ii · 2 marks

2020 JAN · Paper 01 · find part a.ii in this mark scheme extract

Number				
8	(a) (i)	X = urethra (1) Y = ureter (1)		2
	(ii)	Any two from: • transports oxygenated blood from the heart (1) • to the body/named organ (1) • maintains high blood pressure (1)		2
	(iii)	Any two from • blood in A/renal artery contains more oxygen/oxygenated (1) • blood in A/renal artery contains more urea (1) • blood in A/renal artery contains more ions (1) • similar concentration of glucose/B renal vein has less glucose (1) • blood in A/renal artery contains less carbon dioxide (1)	Allow reverse argument for B/renal vein	2
	(b) (i)	Any four from: • dialysis fluid/dialysate contains a low(er) concentration/no urea/waste products/ions (1) • urea/waste/ions diffuse from blood (1) • through dialysis membrane (1) • from high(er) concentration (in the blood) to a low(er) concentration (in the dialysate)/down a concentration gradient (1) • dialysis membrane is partially-permeable(1)	ORA allow selectively permeable membrane	4
	(ii)	• to prevent blood from clotting whilst in the machine (1) • (added as it enters the machine) to prevent clots from blocking the machine (as blood flows through) (1) • (not added at the end) as blood clotting properties need to return to normal before passing back into patient (1)		3

Total for question 8 = 13 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 8 of 17 · Question 5 part b · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

5 (a)	• ACE inhibitor prevents/inhibits action of enzyme; • prevents formation of angiotensin II; • less(angiotensin ii) bind with receptors; • Losartan blocks/binds to receptors; • angiotensin ii cannot bind to receptor;		4
(b)	• arteries/arterioles; • muscles in walls; • can contract/constrict; • narrow lumen; • smaller volume/space(for blood flow);		4
(c)	• in normal/healthy kidney protein not found in urine; • protein is a big molecule; • should not pass through the glomerulus/basement membrane; • damage to glomerulus/basement membrane;		2
(d)	• stronger/intense/eq; <u>contractions</u>; • (which causes) <u>muscle</u> to get thicker;	mp2 must be due to mp1	2
(e)	• detoxification occurs in the liver; • can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver; • results in (further) liver damage/liver failure/liver diseae;		2 **Total =14**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 8 of 17 · Question 5 part b · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks
	• A=rods/cones/retina;		

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 9 of 17 · Question 1 part a.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

1 (a)	(i)	long (bone)		1
	(ii)	<table><tr><th rowspan="2">Part</th><th colspan="3">Function</th></tr><tr><th>reduces friction</th><th>produces red blood cells</th><th>provides strength</th></tr><tr><td>compact bone</td><td></td><td></td><td>✓</td></tr><tr><td>spongy bone</td><td></td><td>✓</td><td></td></tr><tr><td>cartilage</td><td>✓</td><td></td><td></td></tr></table>		3
	(iii)	• provides nutrients/named nutrient/oxygen; • removes waste/named waste		2
(b)	(i)	In the following order: • tendons; • contracts; • relaxes; • pulls;		4
	(ii)	0.5 x 125; 62.5 (J);	Accept 63 (J)	2
			Total for Question 1 = 12 marks	

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 10 of 17 · Question 1 part b.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

1 (a) (i)	• transport oxygen; • from lungs; • to (body) tissues/cells;	Allow organs	3
(ii)	• haemoglobin to carry/combine with oxygen; • no nucleus, so more haemoglobin; • biconcave shape to increase surface area; • small/flexible (cell) membrane to pass through capillaries; • thin membrane to reduce diffusion distance;		4
(b) (i)	vein;		1
(ii)	• valve present/large lumen/thin (muscular) walls; • only found in veins/not found in arteries/other blood vessels;		2
(iii)	• distance 13-15 mm; • divide by 10 = 1.3-1.5;	Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct	2 Total 12

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 10 of 17 · Question 1 part b.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 11 of 17 · Question 1 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

1 (a) (i)	• transport oxygen; • from lungs; • to (body) tissues/cells;	Allow organs	3
(ii)	• haemoglobin to carry/combine with oxygen; • no nucleus, so more haemoglobin; • biconcave shape to increase surface area; • small/flexible (cell) membrane to pass through capillaries; • thin membrane to reduce diffusion distance;		4
(b) (i)	vein;		1
(ii)	• valve present/large lumen/thin (muscular) walls; • only found in veins/not found in arteries/other blood vessels;		2
(iii)	• distance 13-15 mm; • divide by 10 = 1.3-1.5;	Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct	2 Total 12

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 11 of 17 · Question 1 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 12 of 17 · Question 3 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a)	(i)	arrow from top to bottom;		1
	(ii)	• A is an artery, thicker muscle wall; • to pump blood from heart/to body; • B is a vein with a valve present; • to control direction of blood flow/to prevent backflow;	If no other marks scored award one for correct identification of both vessels	4
	(iii)	capillary;		1
(b)	(i)	• decrease in deaths; • in both sexes/males and females; • between 2005 - 2014/over whole range;		3
	(ii)	• 60 000 000 ÷ 100 000 = 600; • 600 × 640; • = 384 000;		3

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 12 of 17 · Question 3 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 13 of 17 · Question 6 part a.i · 4 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

6 (a)	(i)	• 1 = artery/ aorta/arterioles/named artery; • 2 = vein/ venules/vena cava; • 3 = capillary; • 4 = vein;		4
	(ii)	• 1 = thick wall/muscular wall/elastic wall/fixed/rigid shape/small lumen; • 2 = thin wall/little muscle/little elastic tissue/ irregular shape/large lumen; • 3 = one cell thick/pores/gaps; • 4 = valves;		4
(b)		• M = (left) ventricle/heart; • N = arteries/arterioles/aorta/named artery; • O = capillaries; • P = veins/venules/vena cava;		4

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 13 of 17 · Question 6 part a.i · 4 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 14 of 17 · Question 6 part a.ii · 4 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

6 (a)	(i)	• 1 = artery/ aorta/arterioles/named artery; • 2 = vein/ venules/vena cava; • 3 = capillary; • 4 = vein;		4
	(ii)	• 1 = thick wall/muscular wall/elastic wall/fixed/rigid shape/small lumen; • 2 = thin wall/little muscle/little elastic tissue/ irregular shape/large lumen; • 3 = one cell thick/pores/gaps; • 4 = valves;		4
(b)		• M = (left) ventricle/heart; • N = arteries/arterioles/aorta/named artery; • O = capillaries; • P = veins/venules/vena cava;		4

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 14 of 17 · Question 6 part a.ii · 4 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 15 of 17 · Question 3 part b · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

3 (a) (i)	C; (red blood cells)		1
(ii)	any three from • more carbon dioxide (1) • less oxygen (1) • moister (1) • warmer (1)		3
(iii)	• large surface area (1) • thin (walled)/wall one cell thick (1) • good blood supply (1) • moist lining (1)		3
(b)	• thin wall (1) • narrow lumen (1)	Allow references to only one blood cell passing through	2
(c)	• breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1) • as lungs would not squeeze out air/forced exhalation(1)		2

Total 11 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 15 of 17 · Question 3 part b · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 16 of 17 · Question 3 part a.i · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a)	(i)	arrow from top to bottom;		1
	(ii)	• A is an artery, thicker muscle wall; • to pump blood from heart/to body; • B is a vein with a valve present; • to control direction of blood flow/to prevent backflow;	If no other marks scored award one for correct identification of both vessels	4
	(iii)	capillary;		1
(b)	(i)	• decrease in deaths; • in both sexes/males and females; • between 2005 - 2014/over whole range;		3
	(ii)	• 60 000 000 ÷ 100 000 = 600; • 600 × 640; • = 384 000;		3

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 16 of 17 · Question 3 part a.i · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 17 of 17 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

3 (a)	(i)	arrow from top to bottom;		1
	(ii)	• A is an artery, thicker muscle wall; • to pump blood from heart/to body; • B is a vein with a valve present; • to control direction of blood flow/to prevent backflow;	If no other marks scored award one for correct identification of both vessels	4
	(iii)	capillary;		1
(b)	(i)	• decrease in deaths; • in both sexes/males and females; • between 2005 - 2014/over whole range;		3
	(ii)	• 60 000 000 ÷ 100 000 = 600; • 600 × 640; • = 384 000;		3

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 17 of 17 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks